Bansilal Ramnath Agarwal Charitable Trust's
VISHWAKARMA INSTITUTE OF TECHNOLOGY – PUNE

Logic Design and Microprocessor

Division	A
Batch	1
GR-no	1251070524
Roll no	2070120
Name	Samiksha Sanjay Adake

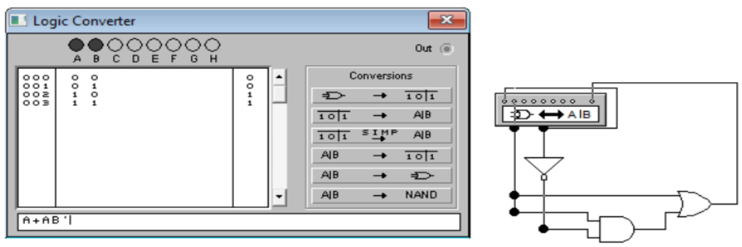
Assignment 1:

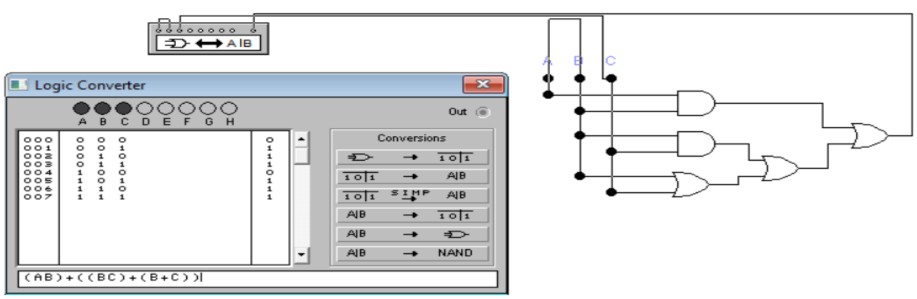
TITLE:- Verification of Logical Gates and Simplification of logic equation using Boolean Algebra

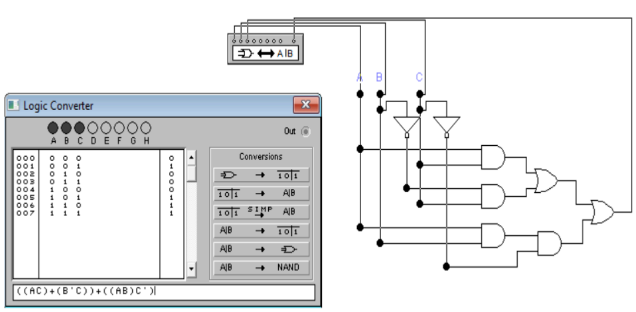
Description:- This assignment presents the implementation of the selected digital logic circuit using appropriate logic gates. It includes the complete circuit design along with its simulation to demonstrate the working of the circuit. The behavior of the circuit is verified by applying different input combinations and observing the corresponding outputs. The assignment also contains the circuit diagram, simulation screenshots, and output results to illustrate the successful operation of the designed logic circuit. The implementation follows the concepts of digital logic and verifies that the

circuit performs the intended logical function correctly.

- **Circuit Diagram**

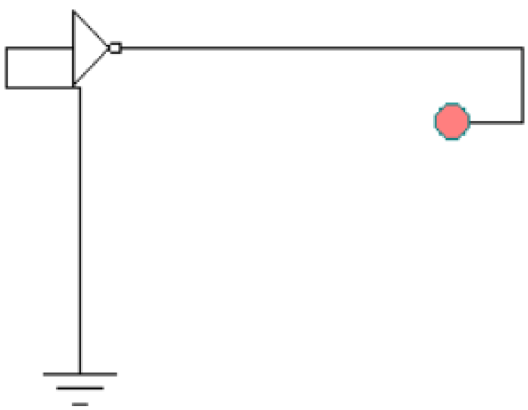


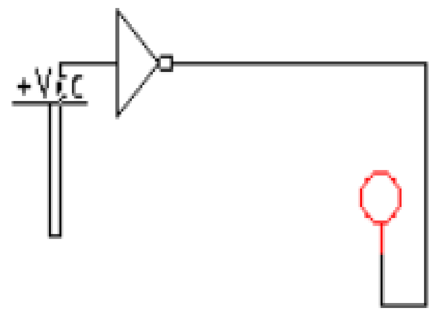




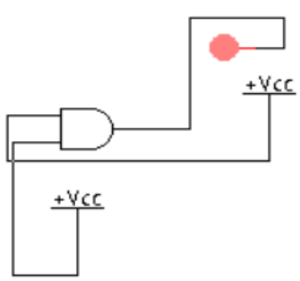
- **Screenshots/Output:**

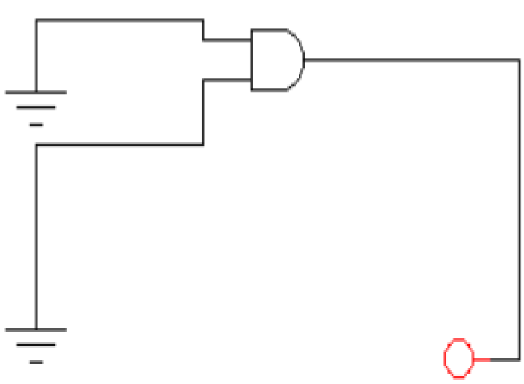
NOT Gate

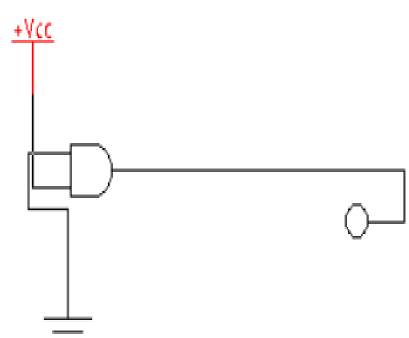




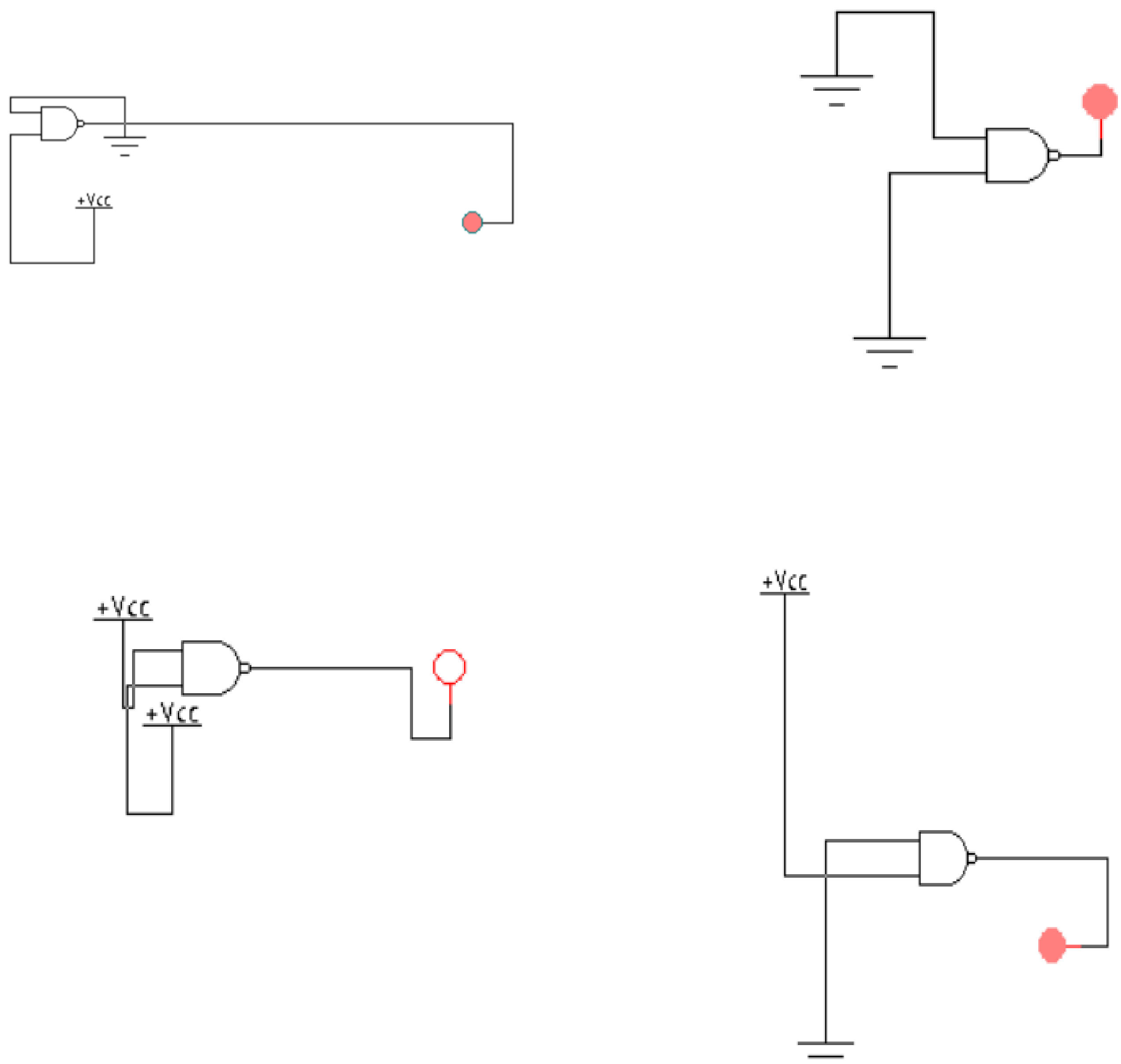
AND Gate



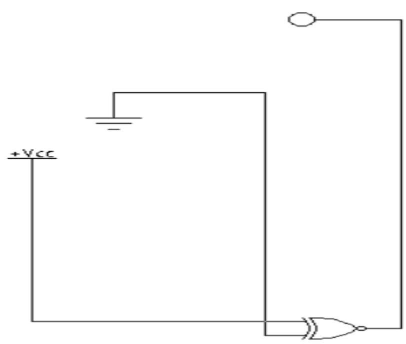


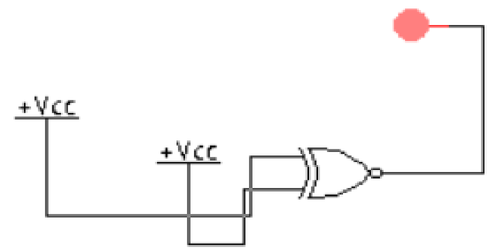


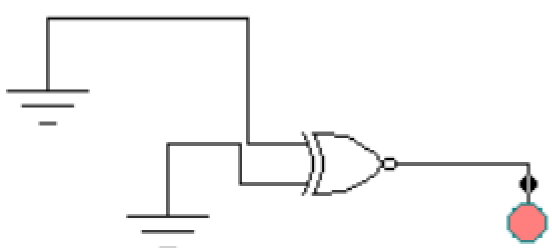
NAND Gate

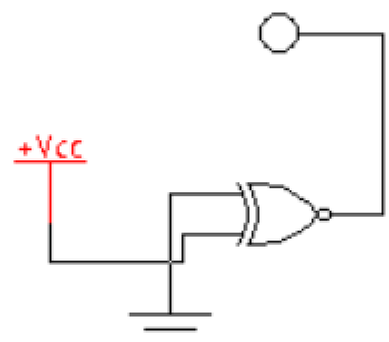


XNOR Gate

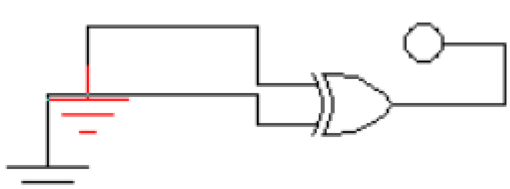


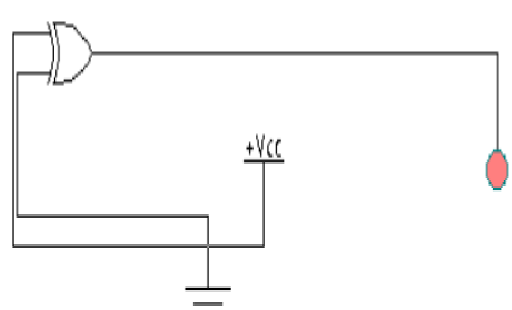


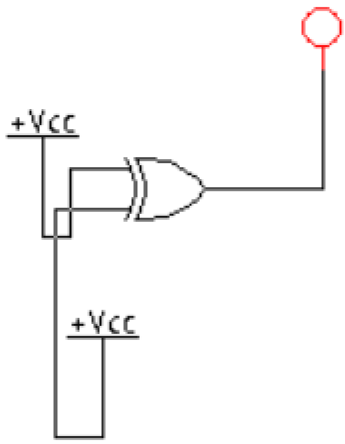


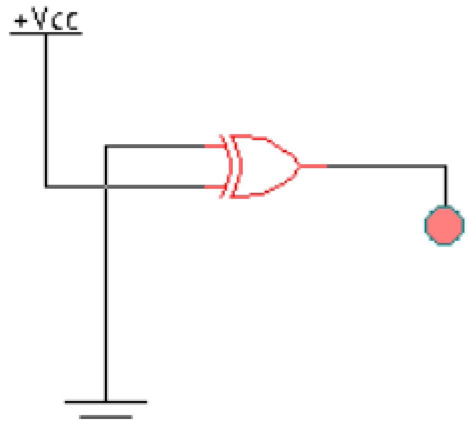


XOR Gate

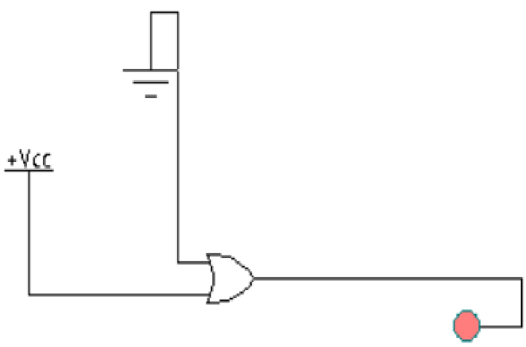


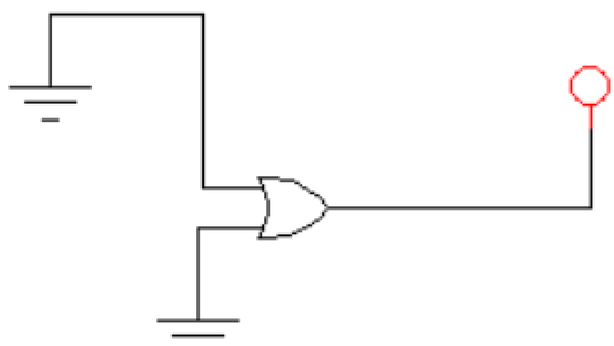


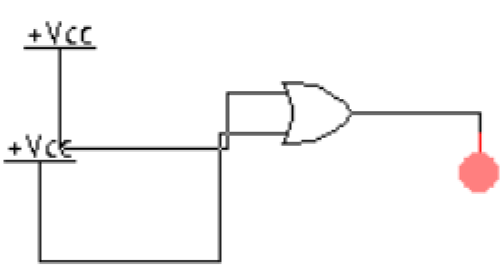


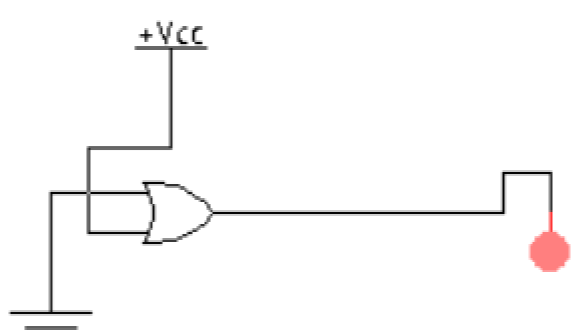


OR Gate

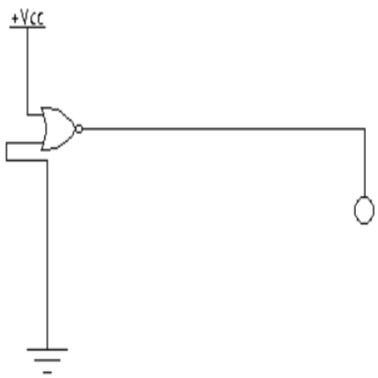


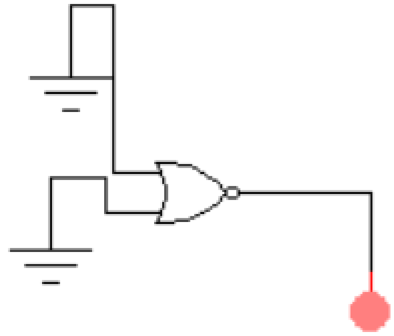


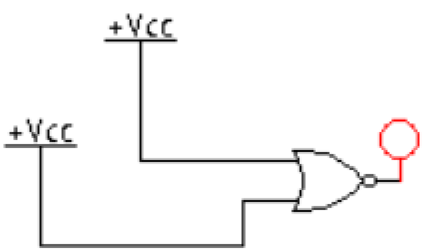


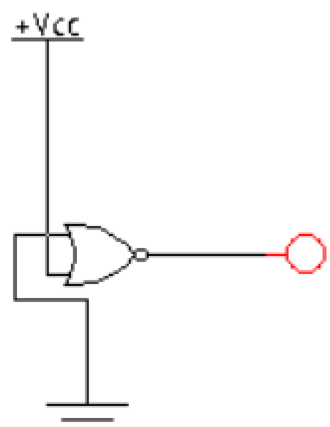


NOR Gate

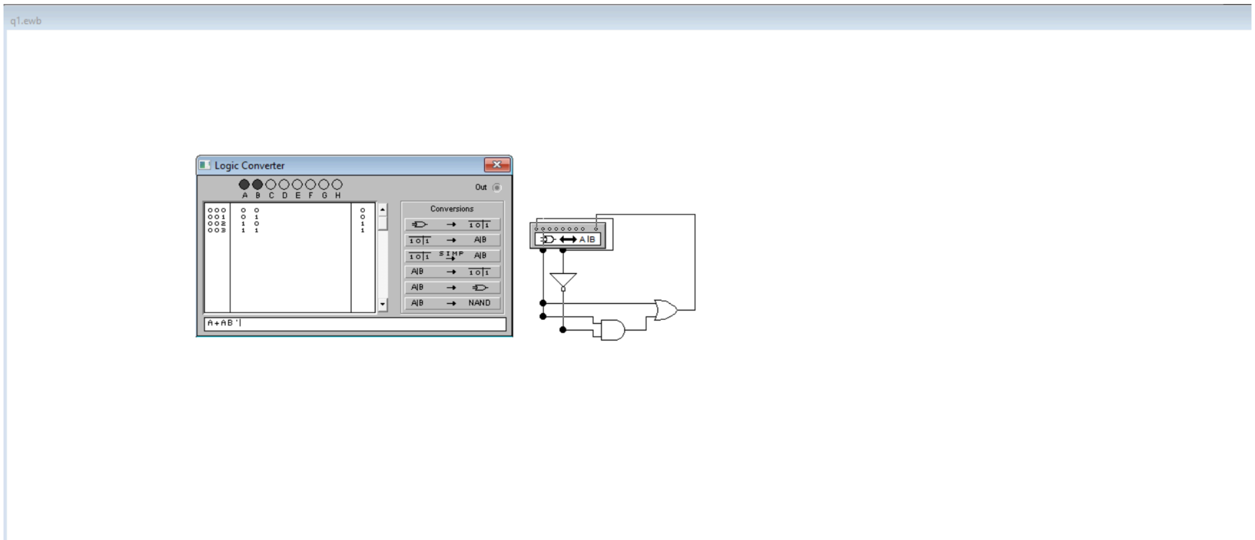




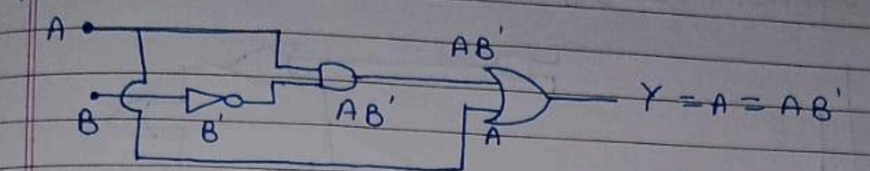




Q1) Y=A+AB’



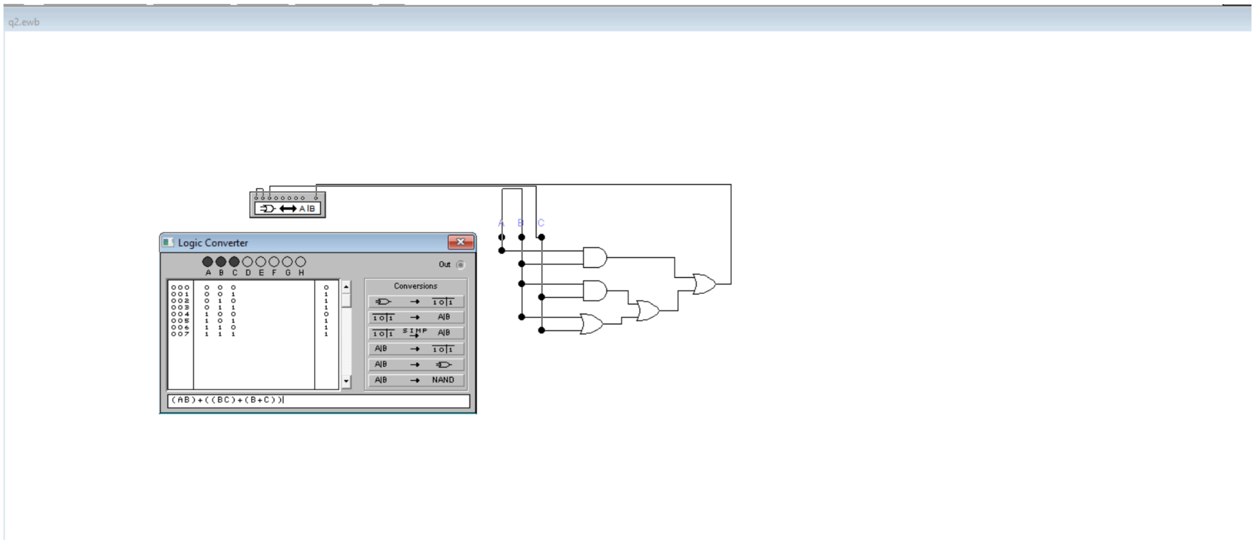
1) Y = A + AB'



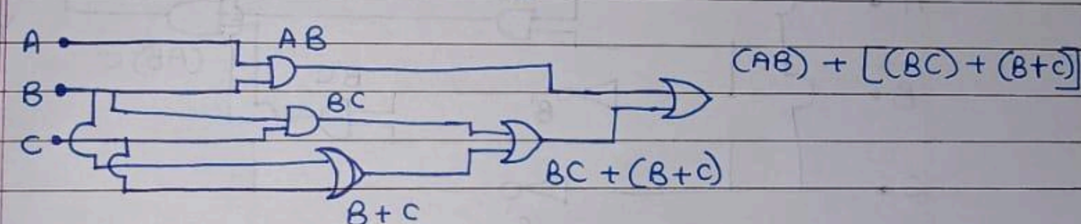
Truth Table

A	B	B'	AB'	A + AB'
0	0	1	0	0
0	1	0	0	0
1	0	1	1	1
1	1	0	0	1

Q2)Y=(AB)+[(BC)+(B+C)]

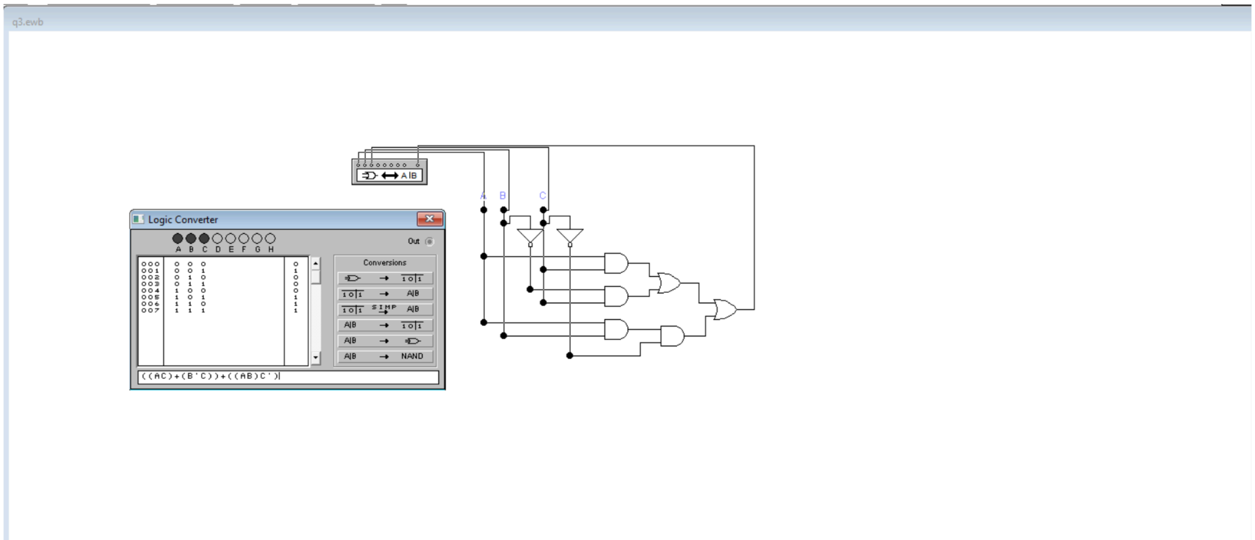


2) Y = (AB) + [(BC) + (B+C)]

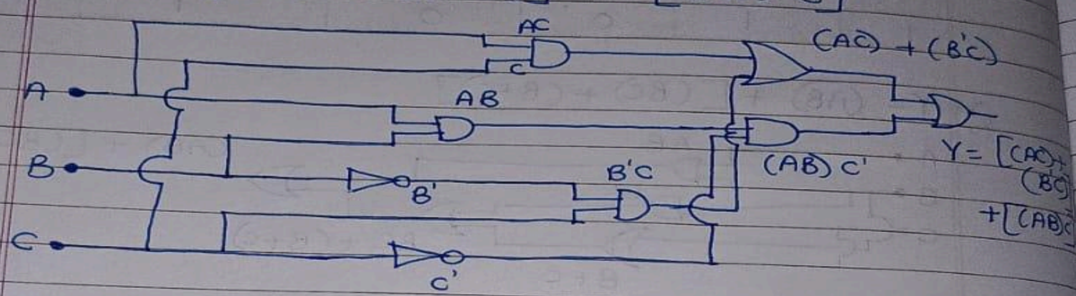


A	B	C	AB	BC	B+C	(BC) + (B+C)	(AB) + [(BC)+(B+C)]
0	0	0	0	0	0	0	0
0	0	1	0	0	1	1	1
0	1	0	0	0	1	1	1
0	1	1	0	1	1	1	1
1	0	0	0	0	0	0	0
1	0	1	0	0	1	1	1
1	1	0	1	0	1	1	1
1	1	1	1	1	1	1	1

Q3)Y=[(AC)+(B’C)]+[(AB).C’]



3) $Y = [(AC) + (\bar{B} \cdot C)] + [(AB) \cdot C']$



A	B	C	B'	C'	AC	B'C	AB	(AB)C'
0	0	0	1	1	0	0	0	0
0	0	1	1	0	0	1	0	0
0	1	0	0	1	0	0	0	0
0	1	1	0	0	0	0	0	0
1	0	0	1	1	0	0	0	0
1	0	1	1	0	1	1	0	0
1	1	0	0	1	0	0	1	1
1	1	1	0	0	1	0	1	0

$[(AC) + (B'C)]$	$[(AC) + (B'C)] + [(AB)C']$
0	0
1	1
0	0
0	0
0	0
1	1
0	1
1	1